

12.421/12.621 Lecture 13: Radiometry and Radiative Transfer

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Topics covered

- Review of black body radiation
- Overview of radiometry
- Radiative transfer equation

1 Radiometry

- Remote sensing method for measuring EM radiation from planetary surfaces, atmospheres, and other natural media
 - More specifically, radiometry is measure of incoherent radiant EM energy

1.1 Review of black body radiation

- To get started, recall Planck's law from last lecture

$$B_f = \frac{2hf^3}{c^2} \left[\frac{1}{e^{hf/k_B T} - 1} \right] \quad \left[\frac{\text{W}}{\text{m}^2 \text{ sr Hz}} \right] \equiv \text{spectral radiance} \quad (1)$$

$$h \equiv \text{Planck's const} = 6.63 \times 10^{-34} \text{ J}\cdot\text{s}$$

$$f = \frac{\omega}{2\pi}$$

$$c \equiv \text{speed of light}$$

$$k_B \equiv \text{Boltzmann's const} = 1.38 \times 10^{-23} \frac{\text{J}}{\text{K}}$$

$$T \equiv \text{absolute temperature}$$

- Recall that when we increase T , we increase the frequency of max B_f (equiv., we decrease the wavelength according to $\lambda_{\text{max}} \sim T^{-1} \rightarrow$ Wien's displacement law) and total radiated power (area under B_f curve)
- How is this used in practice?
 - ↪ First note that B_f is the power emitted by a black body with a 1 m^2 surface area through a 1 sr solid angle over a bandwidth of 1 Hz (area projected normal to the direction of emission, which is why the flux into a hemisphere is πB_f rather than $2\pi B_f$)

↪ Suppose black body has area A_s , the spectral power P_f received by an antenna w/ area A_{eff} at a distance R from black-body is

$$P_f = B_f A_s \Omega_r \quad \left[\frac{\text{W}}{\text{Hz}} \right] \quad (2)$$

where $\Omega_r = \frac{A_{\text{eff}}}{R^2}$ is solid angle subtended by the antenna

↪ Now note that source area subtends a solid angle $\Omega_s = \frac{A_s}{R^2}$

$$\Rightarrow R^2 = \frac{A_{\text{eff}}}{\Omega_r} = \frac{A_s}{\Omega_s}$$

and

$$P_f = B_f A_{\text{eff}} \Omega_s \quad (3)$$

↪ Total power received by a receiver w/ bandwidth $\Delta f = f_{\text{high}} - f_{\text{low}}$

$$P = \int_{f_{\text{low}}}^{f_{\text{high}}} P_f df = A_{\text{eff}} \Omega_s \int_{f_{\text{low}}}^{f_{\text{high}}} B_f df \quad (4)$$

↪ For sufficiently large Δf and an appropriate center frequency, $P/(A_{\text{eff}} \Omega_s) \approx$ total radiance L of black body, and the intensity $I = \pi L$ is given by Stefan–Boltzmann Law (see Lecture 12)

$$I \approx \frac{\pi P}{A_{\text{eff}} \Omega_s} = \sigma_{\text{SB}} T^4 \quad (5)$$

(non-polarized, i.e., dual-polarization, antenna, $c_p = 1$ in Eq. (8); a single-polarization antenna receives half)

where

$$\sigma_{\text{SB}} \equiv \text{Stefan–Boltzmann const} = 5.67 \times 10^{-8} \left[\frac{\text{W}}{\text{m}^2 \text{K}^4} \right]$$

★ But be careful, some RS instruments have narrow bandwidths $\Rightarrow$ Stefan–Boltzmann may not apply

• For relatively narrow bands of most RS instruments, must consider Planck's law or an approximation of it

→ at infrared frequencies, it may be advisable to use Planck's law

→ at microwave frequencies can use Rayleigh–Jeans law

$$B_f = \frac{2k_B T f^2}{c^2} \quad (6)$$

$$\left[\begin{array}{l} \text{To get Rayleigh–Jeans from Planck's consider identity} \\ e^z - 1 = \left(1 + z + \frac{z^2}{2!} + \dots \right) - 1 \quad \left(\text{where } z = \frac{hf}{k_B T} \right) \\ \approx z \quad \text{for } z \ll 1 \rightarrow \text{low frequency limit for which Rayleigh–Jeans applies} \end{array} \right]$$

1.2 Antenna directionality and the power–temperature correspondence

- To understand relationship between power received by radiometer and the temperature of source body, we must account for directionality of antenna that has a radiation pattern represented by an arbitrary function $F(\theta, \phi)$ where θ is the colatitude and ϕ the longitude (spherical coordinates)

$F(\theta, \phi)$ represents the normalized radiation intensity (the antenna gain pattern normalized to a maximum of 1) as a function of direction

→ In the case of radiometry, $F(\theta, \phi)$ indicates the antenna's ability to convert EM radiation coming from direction defined by θ and ϕ into electrical power (signal)

- Most RS antenna will have a lobe structure to the radiation pattern $\Rightarrow$ observational geometry (Figure 1)

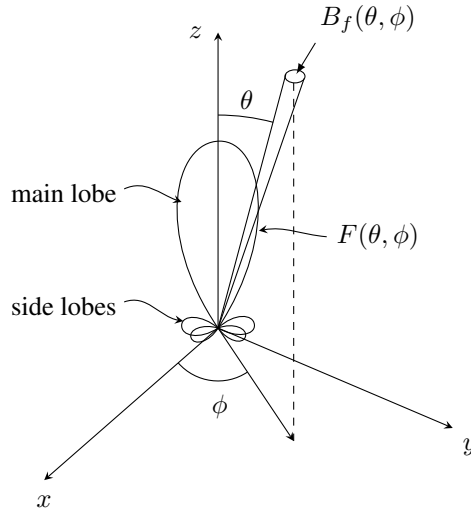


Figure 1: Observational geometry of a radiometer antenna with radiation pattern $F(\theta, \phi)$ (main lobe and side lobes) receiving spectral radiance $B_f(\theta, \phi)$ from the direction (θ, ϕ) (cf. Ulaby and Long (2014), Fig. 6-4).

- To account for antenna directionality let's consider the differential spectral power

$$dP_f = B_f A_{\text{eff}} F(\theta, \phi) d\Omega \quad (7)$$

Then the total power received by antenna over bandwidth $\Delta f = f_2 - f_1$

$$P = c_p A_{\text{eff}} \int_{f_1}^{f_2} \iint_{4\pi} B_f F(\theta, \phi) d\Omega df \quad (8)$$

where c_p accounts for polarization of antenna

$$c_p = \begin{cases} 1 & \text{non-polarized antenna} \\ \frac{1}{2} & \text{polarized antenna} \end{cases}$$

Why does $c_p = \frac{1}{2}$ for polarized antenna?

Because EM wave can be represented by two orthogonal polarizations, each w/ half of the total power. (Think: polarized sunglasses) Black bodies radiate unpolarized EM energy $\Rightarrow$ polarizing filter cuts total power in half

- Now assuming Rayleigh–Jeans law is a valid approximation (which is true for microwaves), total power received from a black-body (antenna inside a black-body enclosure at uniform T ; Ulaby and Long, 2014, Fig. 6-5a)

$$P_{\text{bb}} = c_p A_{\text{eff}} \int_{f_1}^{f_2} \iint_{4\pi} \frac{2k_B T}{\lambda^2} F(\theta, \phi) d\Omega df \quad (9)$$

- Taking the common case in RS of a narrow bandwidth where $B_f \approx \text{constant over } \Delta f$

$$P_{\text{bb}} = k_B T \Delta f \underbrace{\frac{A_{\text{eff}}}{\lambda^2} \iint_{4\pi} F(\theta, \phi) d\Omega}_{= \Omega_p \equiv \text{antenna pattern solid angle}} \quad \text{for } c_p = \frac{1}{2} \quad (10)$$

- From antenna theory (Ulaby and Long, 2014, ch. 3)

$$\Omega_p = \frac{\lambda^2}{A_{\text{eff}}} \quad (11)$$

$$\Rightarrow \boxed{P_{\text{bb}} = k_B T \Delta f} \quad (12)$$

$\rightarrow$ power received at antenna is proportional to T w/ const of proportionality $= k_B \Delta f$ (one a universal const the other an instrument parameter, both known)

$\Rightarrow$ common in radiometry to interchange terms ‘power’ and ‘temperature’

1.3 Brightness temperature and emissivity

- Now, a black-body is an idealized material; Real materials emit less radiation than black bodies (called gray bodies; white bodies reflect all incident radiation)
- And in RS applications, it is often the case that observed EM radiation has propagated through some medium (e.g., we may be interested in the T of a fire as seen from a satellite) that absorbs and/or scatters EM radiation
- Let’s take each of these factors in kind
- Let us consider a semi-infinite material w/ brightness (intensity) $I(\theta, \phi)$ and physical (or actual) temperature T . The observed brightness

$$\begin{aligned} I(\theta, \phi) &= e(\theta, \phi) \frac{P_{\text{bb}}}{\lambda^2} \\ &= \frac{k_B}{\lambda^2} \Delta f e(\theta, \phi) T = \frac{k_B}{\lambda^2} \Delta f T_B(\theta, \phi) \end{aligned} \quad (13)$$

(for one polarization)

where

$T_B = eT \equiv$ brightness temperature (homogeneous, isothermal medium; Ulaby and Long, 2014, p. 235)

$e(\theta, \phi) \equiv$ emissivity, defines material’s ability to emit EM radiation

- In terms of power (per unit area), emissivity is

$$e(\theta, \phi) = \frac{I(\theta, \phi)}{I_{\text{bb}}} = \frac{P(\theta, \phi)}{P_{\text{bb}}} \quad (14)$$

$$\Rightarrow 0 \leq e(\theta, \phi) \leq 1$$

$\uparrow$ white body (nothing absorbed $\Rightarrow$ nothing emitted) $\uparrow$ black-body (total absorption and emission)

- Brightness temperature often used as a proxy for observed brightness and (equiv.) power because of the proportional relation w/ a known constant of proportionality (in the Rayleigh–Jeans regime, Eq. (13))
- Let us relate T_B to the physical and dielectric properties of the material and atmosphere

2 Radiative transfer

- As we have been discussing, interaction of EM radiation w/ matter involves emission (source) and absorption + scattering (extinction; sink)
- Radiative transfer deals w/ these different interactions

2.1 Extinction and emission in an infinitesimal cylinder

- To derive a simple function for radiative transfer, let us consider EM energy propagating in direction $\hat{\mathbf{R}}$ through an infinitesimal cylinder w/ cross-sectional area dA , length dR and position given by vector $\mathbf{r}$ (Figure 2)

$$I(\mathbf{r}, \hat{\mathbf{R}}) \equiv \text{incident brightness at position } \mathbf{r} \text{ in direction } \hat{\mathbf{R}} \left[\frac{\text{W}}{\text{m}^2 \text{ sr}} \right]$$

$$I(\mathbf{r} + dR \hat{\mathbf{R}}, \hat{\mathbf{R}}) \equiv \text{brightness of EM radiation leaving cylinder} \left[\frac{\text{W}}{\text{m}^2 \text{ sr}} \right]$$

$$I(\mathbf{r}, \hat{\mathbf{R}}') \equiv \text{brightness of EM radiation scattered into the cylinder from surrounding area}$$

- Power (brightness) leaving cylinder is < that entering cylinder due to extinction (scattering + absorption)
- The change from incident brightness $I(\mathbf{r}, \hat{\mathbf{R}})$ due to extinction is

$$dI_{\text{extinction}} = \kappa_e I(\mathbf{r}, \hat{\mathbf{R}}) dR \quad (15)$$

where

$$\begin{aligned} \kappa_e &\equiv \text{extinction coef (introduced in Lecture 9)} \\ &= \kappa_a + \kappa_s \end{aligned}$$

κ_s : $\hookrightarrow$ scattering coef

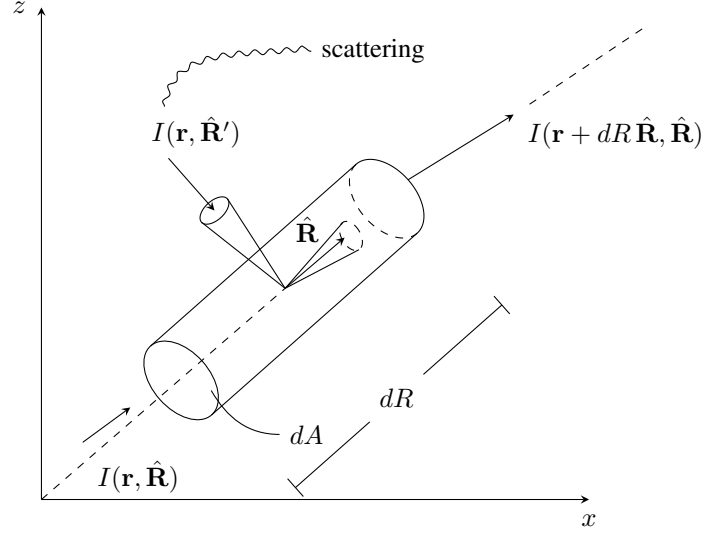


Figure 2: Radiative transfer through an infinitesimal cylinder of cross-sectional area dA and length dR at position $\mathbf{r}$, for propagation in direction $\hat{\mathbf{R}}$; radiation incident from direction $\hat{\mathbf{R}}'$ is scattered into $\hat{\mathbf{R}}$ (cf. Ulaby and Long (2014), Fig. 6-12).

$$\kappa_a: \hookrightarrow \text{absorption coef} = -2k_0 \text{Im}\{\sqrt{\varepsilon}\}$$

- Media inside cylinder emits EM radiation w/ brightness

$$dI_{\text{emission}} = \kappa_e J(\mathbf{r}, \hat{\mathbf{R}}) dR \quad (16)$$

where

$J \equiv$ total source function

- Total source function J is a weighted sum of scattering J_s and absorption (emission) J_a source functions

$$J = (1 - a)J_a + aJ_s \quad (17)$$

where

$$a = \frac{\kappa_s}{\kappa_e} \equiv \text{single-scatter albedo}$$

$J_s \equiv$ scattering source function: accounts for radiation scattered along $\hat{\mathbf{R}}$ (by matter in cylinder) and sourced from EM radiation coming in from all directions (e.g., sunlight scattered by surrounding atmosphere)

$J_a \equiv$ absorption source function: emission must balance absorption when matter is in thermal equilibrium (Kirchhoff's Law of thermal radiation) $\rightarrow$ In RS, this is a reasonable approx. under local thermodynamic equilibrium (collision-dominated; in Earth's atmosphere below ~ 60 – 70 km; Liou, 2002, pp. 14, 27)

2.2 Radiative transfer equation

- Now, propagation over distance dR results in change of (power) intensity dI

$$\begin{aligned}
 dI &= dI_{\text{emission}}^{\text{source}} - dI_{\text{extinction}}^{\text{sink}} \\
 &= \kappa_e J dR - \kappa_e I dR \\
 &= \kappa_e dR \left[J(\mathbf{r}, \hat{\mathbf{R}}) - I(\mathbf{r}, \hat{\mathbf{R}}) \right]
 \end{aligned} \tag{18}$$

- Defining optical depth $d\tau_{\text{opt}} = \kappa_e dR$, we can rearrange to get the radiative transfer equation

$$\boxed{\frac{dI}{d\tau_{\text{opt}}} + I = J} \tag{19}$$

In words: Change in brightness w/ optical depth (product of κ_e and dR) = total source less incident brightness temperature

$\hookrightarrow$ In essence, statement of conservation of energy

2.3 Radiative transfer in terms of brightness temperature

- Relating to brightness temperature T_B

Recall Rayleigh–Jeans

$$I = \frac{k_B}{\lambda^2} \Delta f T_B \tag{20}$$

By Kirchhoff's law, emission = absorption

$$\Rightarrow J_a = \frac{k_B}{\lambda^2} \Delta f T \quad T = T(\mathbf{r}) \tag{21}$$

$\hookrightarrow J_a$ is isotropic $\Rightarrow J_a = J_a(\mathbf{r})$

Analogously, define volume scattered radiometric temperature $T_{VS}(\mathbf{r}, \hat{\mathbf{R}})$ such that

$$J_s(\mathbf{r}, \hat{\mathbf{R}}) = \frac{k_B}{\lambda^2} \Delta f T_{VS}(\mathbf{r}, \hat{\mathbf{R}}) \tag{22}$$

Plugging into radiative transfer equation

$$\boxed{\frac{dT_B}{d\tau_{\text{opt}}} = (1 - a)T + aT_{VS} - T_B} \tag{23}$$

Variables

Units are SI. Values are given for universal constants. The brightness I (from Section 1.3 on) is the brightness of one polarization integrated over the bandwidth Δf (Rayleigh–Jeans form, Eq. (13)).

Symbol	Description	Units	Value
<i>Constants</i>			
c	speed of light in a vacuum	m/s	2.998×10^8
h	Planck constant	J s	6.63×10^{-34}
k_B	Boltzmann constant	J/K	1.38×10^{-23}
σ_{SB}	Stefan–Boltzmann constant	$\text{W m}^{-2} \text{K}^{-4}$	5.67×10^{-8}
<i>Black body radiation and received power</i>			
T	(absolute, physical) temperature	K	–
f	frequency, $\omega/2\pi$	Hz	–
ω	angular frequency	rad/s	–
λ	wavelength	m	c/f
B_f	spectral radiance (per unit frequency)	$\text{W m}^{-2} \text{sr}^{-1} \text{Hz}^{-1}$	–
B_λ	spectral radiance (per unit wavelength)	$\text{W m}^{-2} \text{sr}^{-1} \text{m}^{-1}$	–
λ_{max}	wavelength of the peak of B_λ (Wien)	m	$\propto T^{-1}$
z	$hf/(k_B T)$	dimensionless	RJ: $z \ll 1$
A_s	area of the black-body source	m^2	–
A_{eff}	(effective) area of the receiving antenna	m^2	λ^2/Ω_p
R	distance between source and antenna	m	–
$\Omega, d\Omega$	solid angle; its differential	sr	–
Ω_r	solid angle subtended by the antenna at the source	sr	A_{eff}/R^2
Ω_s	solid angle subtended by the source at the antenna	sr	A_s/R^2
P_f	spectral power received	W/Hz	–
P	total power received	W	–
$f_{\text{low}}, f_{\text{high}}$	lower and upper limits of the receiver band (also f_1, f_2)	Hz	–
Δf	bandwidth	Hz	$f_{\text{high}} - f_{\text{low}}$
L	total radiance of a black body (Lecture 12)	$\text{W m}^{-2} \text{sr}^{-1}$	$\sigma_{\text{SB}} T^4/\pi$
I	(Eq. 5) intensity (flux density) of a black body, πL	W/m^2	$\sigma_{\text{SB}} T^4$
<i>Antenna</i>			
$F(\theta, \phi)$	(normalized) radiation pattern of the antenna	dimensionless	max. 1
θ, ϕ	colatitude and longitude (spherical coordinates)	rad	–
x, y, z	Cartesian coordinates (figures)	m	–
c_p	polarization factor of the antenna	dimensionless	1 (non-polarized), $\frac{1}{2}$ (polarized)
Ω_p	antenna pattern solid angle, $\iint_{4\pi} F d\Omega$	sr	λ^2/A_{eff}
P_{bb}	power received from a black body	W	$k_B T \Delta f (c_p = \frac{1}{2})$

Symbol	Description	Units	Value
<i>Brightness and emissivity</i>			
$I(\theta, \phi)$	brightness (intensity) of a material (one polarization)	$\text{W m}^{-2}\text{sr}^{-1}$	$k_B \Delta f T_B / \lambda^2$
I_{bb}	brightness of a black body	$\text{W m}^{-2}\text{sr}^{-1}$	$k_B \Delta f T / \lambda^2$
$P(\theta, \phi)$	power received from a material	W	–
$e(\theta, \phi)$	emissivity	dimensionless	0–1
T_B	brightness temperature	K	eT
<i>Radiative transfer</i>			
$\mathbf{r}$	position vector	m	–
$\hat{\mathbf{R}}, \hat{\mathbf{R}}'$	direction of propagation; direction of radiation scattered into $\hat{\mathbf{R}}$	dimensionless	–
dA, dR	cross-sectional area and length of the cylinder	$\text{m}^2; \text{m}$	–
$I(\mathbf{r}, \hat{\mathbf{R}})$	brightness at $\mathbf{r}$ in direction $\hat{\mathbf{R}}$	$\text{W m}^{-2}\text{sr}^{-1}$	–
$dI_{\text{extinction}}$	loss of brightness by extinction over dR	$\text{W m}^{-2}\text{sr}^{-1}$	$\kappa_e I dR$
dI_{emission}	gain of brightness by emission over dR	$\text{W m}^{-2}\text{sr}^{-1}$	$\kappa_e J dR$
κ_e	extinction coefficient	Np/m	$\kappa_a + \kappa_s$
κ_a	absorption coefficient	Np/m	$-2k_0 \text{Im}\{\sqrt{\varepsilon}\}$
κ_s	scattering coefficient	Np/m	–
k_0	free-space wavenumber	rad/m	$2\pi/\lambda$
ε	complex dielectric constant	dimensionless	–
J	total source function	$\text{W m}^{-2}\text{sr}^{-1}$	$(1 - a)J_a + aJ_s$
J_a	absorption source function	$\text{W m}^{-2}\text{sr}^{-1}$	$k_B \Delta f T / \lambda^2$
J_s	scattering source function	$\text{W m}^{-2}\text{sr}^{-1}$	$k_B \Delta f T_{\text{VS}} / \lambda^2$
a	single-scattering albedo	dimensionless	κ_s / κ_e
τ_{opt}	optical depth, $\int \kappa_e dR$	Np (dimensionless)	–
T_{VS}	volume scattered radiometric temperature	K	–